Flow-Based Construction for Auditable Redistricting Assignments

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Abstract

This paper presents T.17, a flow-based construction baseline for BISECT. The paper focuses on capacity and cost structure, deterministic fixture behavior, infeasibility witnesses, and audit-compatible summaries. The staged implementation is intentionally modest: it establishes the contract for flow-family constructors before larger solvers are introduced, and it does not claim global optimality, legal compliance, or political fairness.

1 Introduction

Flow formulations are a natural way to express capacity and assignment costs. T.17 adds a small deterministic flow-construction baseline that records validity and infeasibility in a form suitable for audit and later solver expansion.

The staged T.17 implementation should be read as a contract-setting constructor, not as the final exact-optimization engine. It uses deterministic seeds and a balanced frontier expansion to produce assignments or model-relative failure evidence. The value of the paper is that it defines the evidence surface later flow, branch-and-cut, and branch-and-price solvers can share: status, capacity bounds, seeds, costs, repair setting, and RPLAN/RCTX lineage.

2 Method

The staged method constructs assignments under population capacity limits. It records whether the implemented model produced a valid result, a result needing repair, or an infeasible-capacity result. When capacity is infeasible, the output carries an explicit witness rather than emitting a final plan.

The atlas framing treats eligible arcs as the candidate objects. A unit can send supply only along arcs allowed by the declared graph/profile, and a selected arc remains valid only while the target bin stays inside its lower and upper capacity bounds. The important failure is therefore concrete: the paper should name the bound that fails and whether the artifact is a different feasible extraction, a model-relative infeasibility witness, or an invalid post-extraction block.

2.1 Reproducible Procedure

The constructor uses the graph and unit attributes supplied by the BISECT state pipeline. The baseline is feasibility- and audit-oriented. It is not yet the branch-and-cut or branch-and-price machinery planned for U.16 and U.17.

1. Select deterministic farthest-point seeds on the adjacency graph.
2. Compute lower, ideal, and upper district-population bounds from the declared tolerance.
3. Grow balanced frontiers from the seeds. A frontier arc is eligible only when assigning that unit keeps the target district at or below the upper capacity bound.
4. Fill remaining unassigned units by nearest seed only after frontier growth stalls, then validate population and connectedness.
5. Mark the result `valid` when assignment and validation checks pass.
6. Mark the result `needs-repair` when extraction exists but fails declared checks, and run the staged small-fixture repair when enabled.
7. Mark the result `infeasible-capacity` when no balanced capacity assignment exists for the declared fixture. Invalid input returns an error before a flow artifact is emitted.

Status	Meaning	Artifact behavior
`valid`	Model produces a validated assignment	Write final sidecars
`needs-repair`	Extracted assignment fails declared checks	Try staged repair or block export
`infeasible-capacity`	Model/profile cannot satisfy capacity	Write model-relative witness
`invalid input`	Input shape is malformed	Return an error before artifact emission

Table 1: Flow-construction status taxonomy. Infeasibility is relative to the declared model and profile.

2.2 Worked Frontier Decision

The current baseline does not solve a general min-cost-flow instance. Its flow-like decision is a deterministic frontier assignment under capacity. If the upper bound is 55, a candidate arc into a district with population 48 is rejected for a unit of population 10, while a lighter unit remains eligible.

3 Implementation

The implementation lives in `bisect-flow` and is exposed through the flow-construction structure mode. Summaries record status, edge cut, population deviation, parameter hash, and algorithm lineage.

District frontier	Candidate unit	Current pop.	Pop. after arc	Decision
A	u_7 pop. 10	48	58	reject: exceeds 55
B	u_8 pop. 6	47	53	accept: within bound

Table 2: T.17 exposes capacity-gated frontier arcs. Rejection is evidence about the declared model, not evidence that all possible districting models fail.

The versioned `FlowSummary` records method, seed method, cost method, repair method, status, population deviation, edge cut, seeds, optional infeasibility witness, and parameter hash. The implemented defaults are farthest-point seeds, an edge-cut cost label, and BFS repair. The witness fields are the failure reason, ideal population, lower population, and upper population. Those fields explain the model profile that failed; they are not a certificate of universal infeasibility.

4 Audit Artifacts

Flow-construction summaries become audit evidence through the RPLAN and RCTX sidecar path: assignments, capacity status, infeasibility witnesses where present, parameter hashes, and algorithm lineage are preserved for verifier checks. Passing the audit profile confirms the recorded contract, not global optimality or legal compliance.

The sidecar contract binds the exported assignment or witness to the declared flow model, costs, capacities, parameters, validation outcome, and algorithm lineage. Parameter hashes are intended to detect drift in the run configuration. An infeasibility witness is model-relative: it explains why the declared flow profile failed, not why no possible redistricting plan can exist.

This distinction is especially important because U.16 and U.17 are the papers that carry exact optimization and bound language. T.17 only promises that the construction run and its status can be replayed and audited under the declared profile.

5 Evaluation Plan

The current evidence layer includes path balance, infeasible capacity, determinism fixtures, and a benchmark-tier RPLAN package generated from the flow constructor on a 100-unit synthetic path. The next evidence layer should characterize real-data scale limits and compare flow construction with clustering, regionalization, and METIS baselines.

The evaluation standard is model-relative. A path fixture can prove that the frontier assignment respects the implemented capacity contract. An infeasibility fixture can prove that a witness is emitted when the declared capacity bounds cannot be met. Neither result proves district quality, legal sufficiency, or optimality against exact U-series solvers.

Claim	Evidence now	Missing evidence	Status
Path fixture balance	L0 path tests	None for fixtures	Current
Infeasible capacity witness	L0 infeasibility tests	Broader case sweep	Current
Deterministic output	L0 determinism tests	Larger graph sweep	Current
Audit lineage	L1 sidecar checks	Real export transcript	Current
Scale smoke	path100 benchmark package	Real-data benchmarks	Benchmark smoke

Table 3: Evidence ladder for T.17 claims. Feasibility, witness generation, and plan quality are separate evidence levels.

Constructor	Capacity behavior	Failure artifact	Current role
Flow construction	model capacity bounds	infeasibility witness	baseline
Capacity clustering	status/repair	witness or repair lineage	comparison
Regionalization	merge constraints	merge/status summary	comparison
Spectral	sweep balance	validation failure	comparison
METIS	external partitioning	manifest/validation result	comparison

Table 4: Failure-behavior comparison framing. Quantitative comparisons remain future work until commands and outputs are recorded.

6 Limitations

The current flow constructor is a baseline, not a mature min-cost-flow solver for every redistricting instance. Capacity feasibility and district quality must be evaluated separately.

The implementation is closer to deterministic capacity-gated frontier growth than to a full network simplex, multi-commodity flow, or branch-and-price solver. That is an acceptable construction baseline, but the paper must not borrow stronger optimization guarantees from algorithms that are not yet part of this slice.

Flow feasibility is also model-relative. A successful flow assignment does not prove compactness, community preservation, partisan fairness, or legal compliance. An infeasibility witness explains failure under the declared graph, costs, capacities, and profile; it is not proof that no valid plan exists under another model.

7 Conclusion

T.17 brings flow construction into the BISECT algorithm family while preserving the shared final-plan audit path used across the T/U roadmap. Its role is a model-relative, auditable construction baseline; stronger exact optimization claims belong to the later U.16 and U.17 papers.

The immediate contribution is therefore the witness contract: a flow-family constructor can say what bounds, seeds, costs, repair mode, and status produced the artifact. That makes later exact solvers easier to compare without overstating what this first construction slice has proven.

References

BISECT Project. T.17 flow-based construction implementation spec, 2026. BISECT internal specification.